

This LM is to add security group module and use it in the parent main.tf

Steps

1. Create the security group module

In the module github, create an “**security-groups**” folder

Filename1: sg.tf

```
# Security Group - base security group (rules defined separately)
```

```
resource "aws_security_group" "web_sg" {  
  name      = "lab9-web-sg-${var.project}"  
  description = "Security group for web server"  
  vpc_id    = var.vpc_id
```

```
  tags = {  
    Name = "lab9-web-sg-${var.project}"  
  }  
}
```

```
# Security Group Rule - HTTP ingress
```

```
resource "aws_vpc_security_group_ingress_rule" "http" {  
  security_group_id = aws_security_group.web_sg.id  
  description       = "HTTP"  
  from_port         = 80  
  to_port           = 80  
  ip_protocol       = "tcp"  
  cidr_ipv4         = "0.0.0.0/0"
```

```
  tags = {  
    Name = "lab9-http-rule-${var.project}"  
  }  
}
```

```
# Security Group Rule - ssh ingress
```

```

resource "aws_vpc_security_group_ingress_rule" "ssh" {
  security_group_id = aws_security_group.web_sg.id
  description      = "ssh"
  from_port        = 22
  to_port          = 22
  ip_protocol       = "tcp"
  cidr_ipv4         = "0.0.0.0/0"

  tags = {
    Name = "lab9-http-rule-${var.project}"
  }
}

# Security Group Rule - Allow all outbound
resource "aws_vpc_security_group_egress_rule" "all_outbound" {
  security_group_id = aws_security_group.web_sg.id
  description       = "Allow all outbound"
  ip_protocol        = "-1"
  cidr_ipv4          = "0.0.0.0/0"

  tags = {
    Name = "lab9-egress-rule-${var.project}"
  }
}

```

Filename2: sg_var.tf

```

variable "vpc_id" {
  description = "VPC ID where security groups will be created"
  type        = string
}

```

```

variable "project" {
    description = "Project name - use UserX format where X is your student number"
    type        = string
}

```

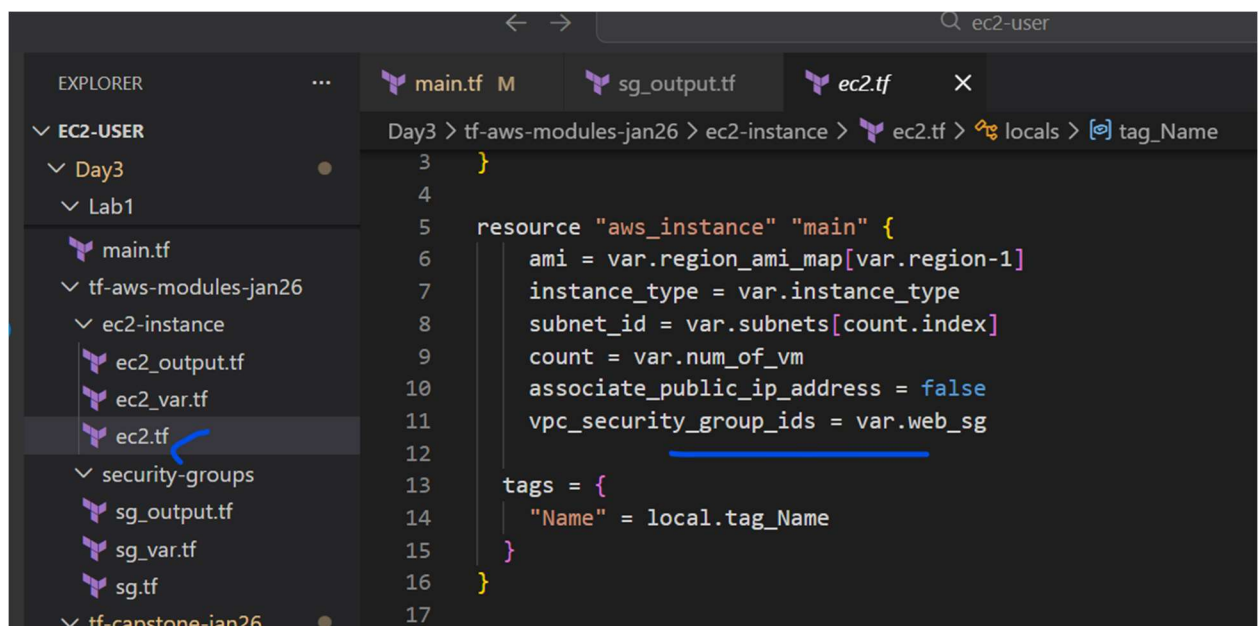
Filename3: sg_output.tf

```

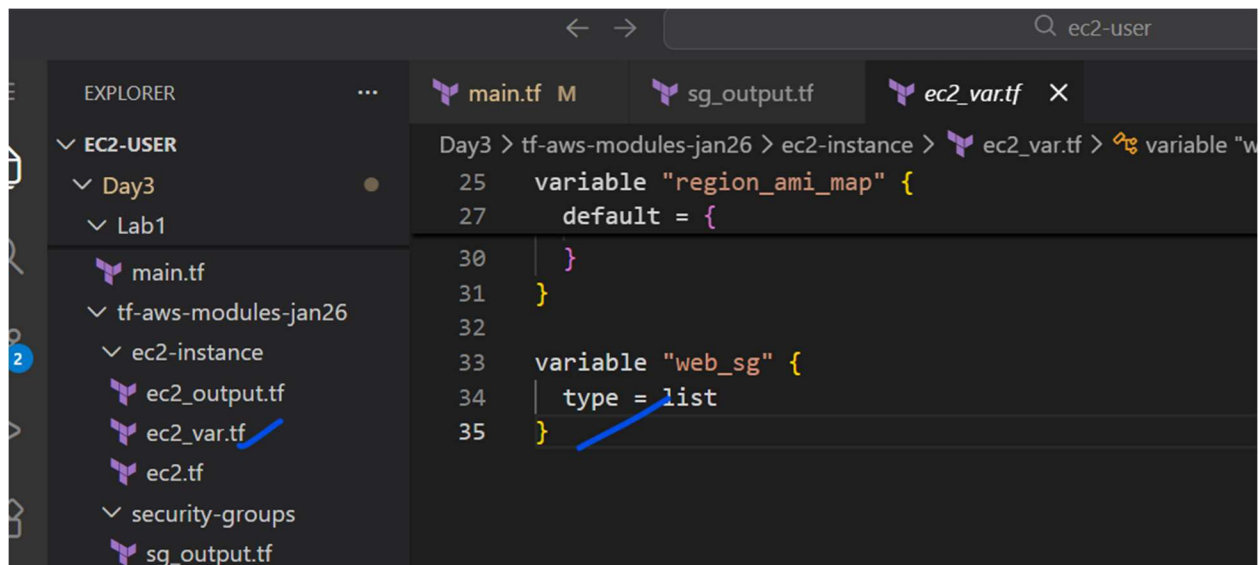
output "web_sg_id" {
    description = "ID of the web security group"
    value       = aws_security_group.web_sg.id
}

```

2. Add security group argument in the “ec2-instance” module

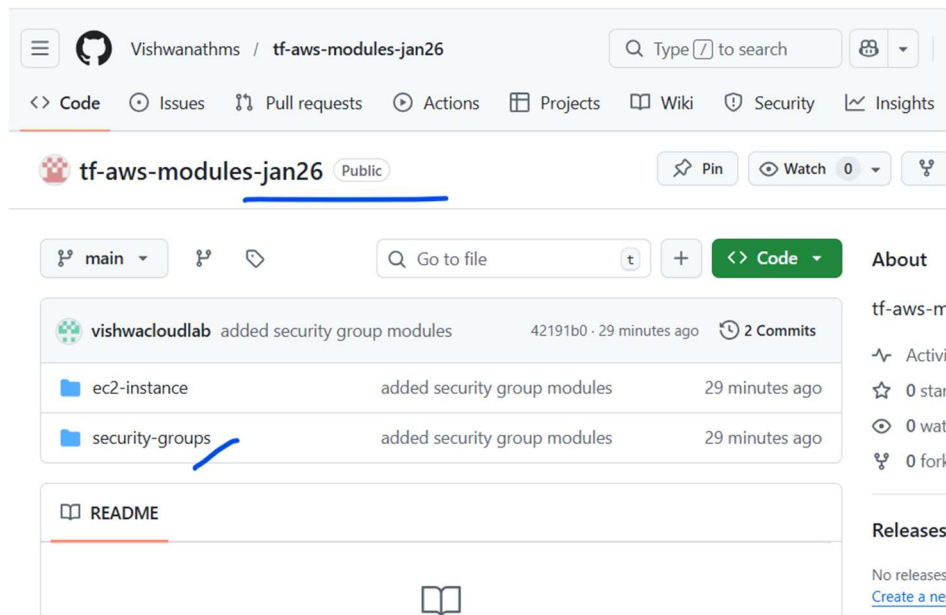


Add the below to the “ec2_var.tf” file



```
variable "web_sg" {
  type = list
}
```

3. Push the changes to the module github repo

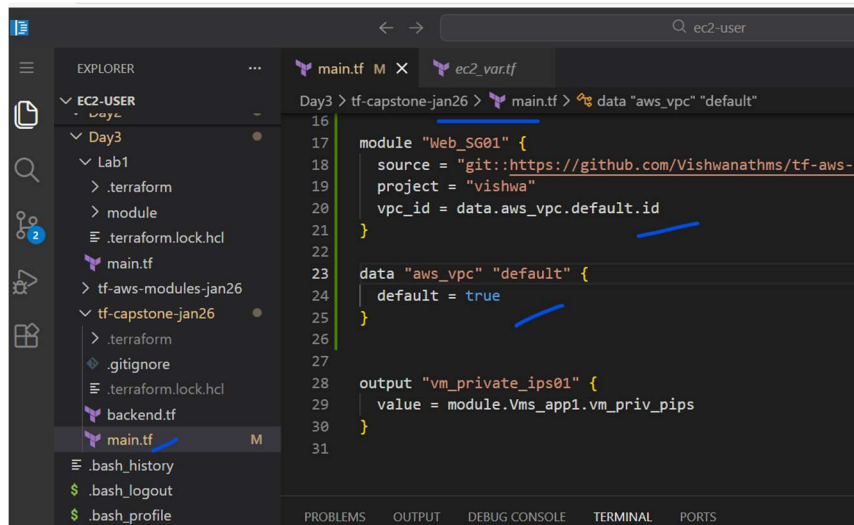


4. Add the module to the main.tf

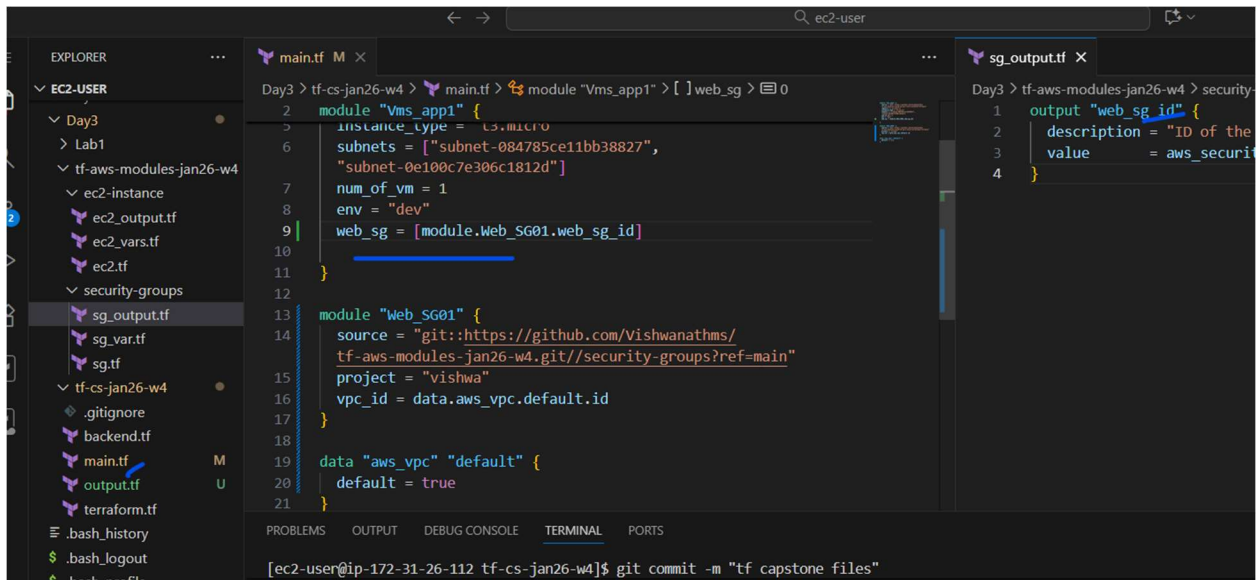
We are using the data to fetch the VPC ID automatically, instead of hard coding.

```
module "Web_SG01" {
  source = "git::https://github.com/Vishwanathms/tf-aws-modules-
jan26.git//security-groups?ref=main"
  project = "vishwa"
  vpc_id = data.aws_vpc.default.id
}

data "aws_vpc" "default" {
  default = true
}
```

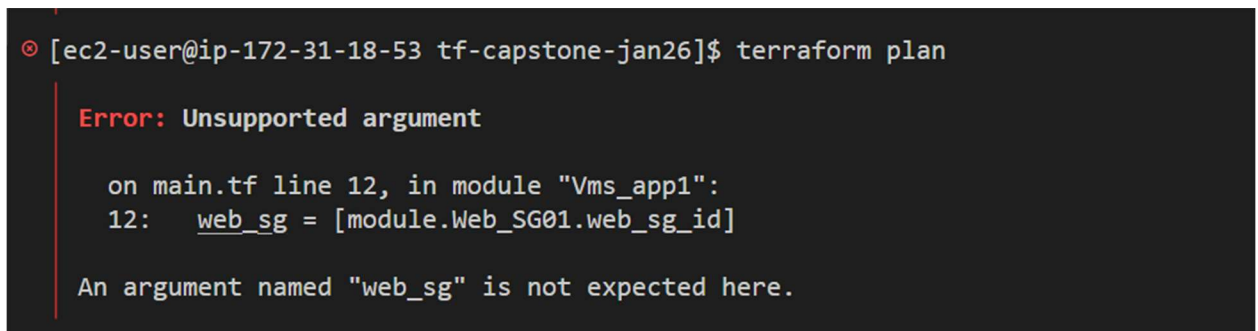


5. Update the "vms_app1" module with "web_sg" variable

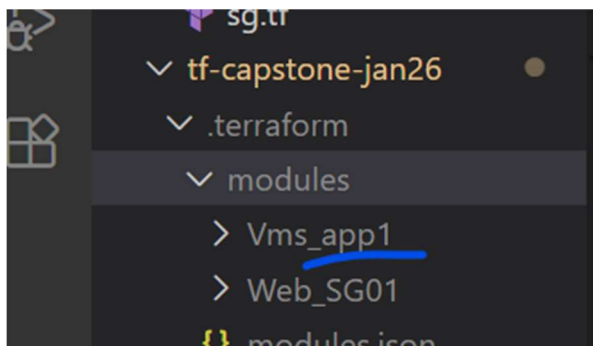


6. Run terraform init and plan

We will get the below error



This is bcz the cached module folder in the .terraform in the capstone has the old module data for "Vms_app1"



Delete that folder "Vms_app1" and run

Terraform init

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

[ec2-user@ip-172-31-18-53 tf-capstone-jan26]$ terraform plan
Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exa
actions if you run "terraform apply" now.
[ec2-user@ip-172-31-18-53 tf-capstone-jan26]$ terraform init
Initializing the backend...
Initializing modules...
Downloading git::https://github.com/Vishwanathms/tf-aws-modules-jan26.git?ref=main for Vms_app1.
- Vms_app1 in .terraform/modules/Vms_app1/ec2-instance
Downloading git::https://github.com/Vishwanathms/tf-aws-modules-jan26.git?ref=main for Web_SG01.
- Web_SG01 in .terraform/modules/Web_SG01/security-groups
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.28.0

Terraform has been successfully initialized!
```

And then the terraform plan and apply will work

Output.

> [Instances](#) > i-010b630bd8d0d5cbf

id

ual View [↗](#)

Types

emplates

uests

Plans

Instances

Hosts

Reservations

Manager [New](#)

630bd8d0d5cbf

Operator

-

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Security details

IAM Role

-

Security groups

[sg-0a8b3b7c56646aa20 \(lab9-web-sg-vishwa\)](#)

▼ Inbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Source
lab9-http-rule-vishwa	sgr-05a52216833c05422	22	TCP	0.0.0.0/0
lab9-http-rule-vishwa	sgr-0615f1645c56085f2	80	TCP	0.0.0.0/0

Owner ID

[992382386705](#)

Launch time

Wed Jan 21 2026 14:29:06 GM